

Human Development Index (HDI) Prediction System Using Machine Learning

Project Description

The **Human Development Index (HDI) Prediction System** is a Machine Learning-based web application developed to predict the Human Development Index (HDI) of a country using important socio-economic indicators. The system helps users estimate the level of human development by analyzing factors such as **Life Expectancy, Mean Years of Schooling, Gross National Income (GNI) per Capita, and Internet Users**.

The application uses the **Linear Regression** machine learning algorithm to predict the HDI score based on historical development data. The backend is developed using **Python** and **Flask**, while data preprocessing and model training are performed using **Pandas, NumPy, and Scikit-learn**. The trained model is stored using **Pickle**, allowing the application to generate predictions efficiently during runtime.

The web application provides a simple and interactive interface where users can enter the required development indicators, submit the data, and instantly receive the predicted HDI score along with its corresponding development category (Low, Medium, High, or Very High). This system can support researchers, students, and policymakers in understanding and analyzing the development status of different countries.

Scenarios

Scenario 1: Government Policy Planning

A government policymaker enters a country's socio-economic indicators such as Life Expectancy, Mean Years of Schooling, GNI per Capita, and Internet Users into the application. The system predicts the country's Human Development Index (HDI) and displays its development category. This helps policymakers evaluate national development and make informed decisions for planning welfare programs and resource allocation.

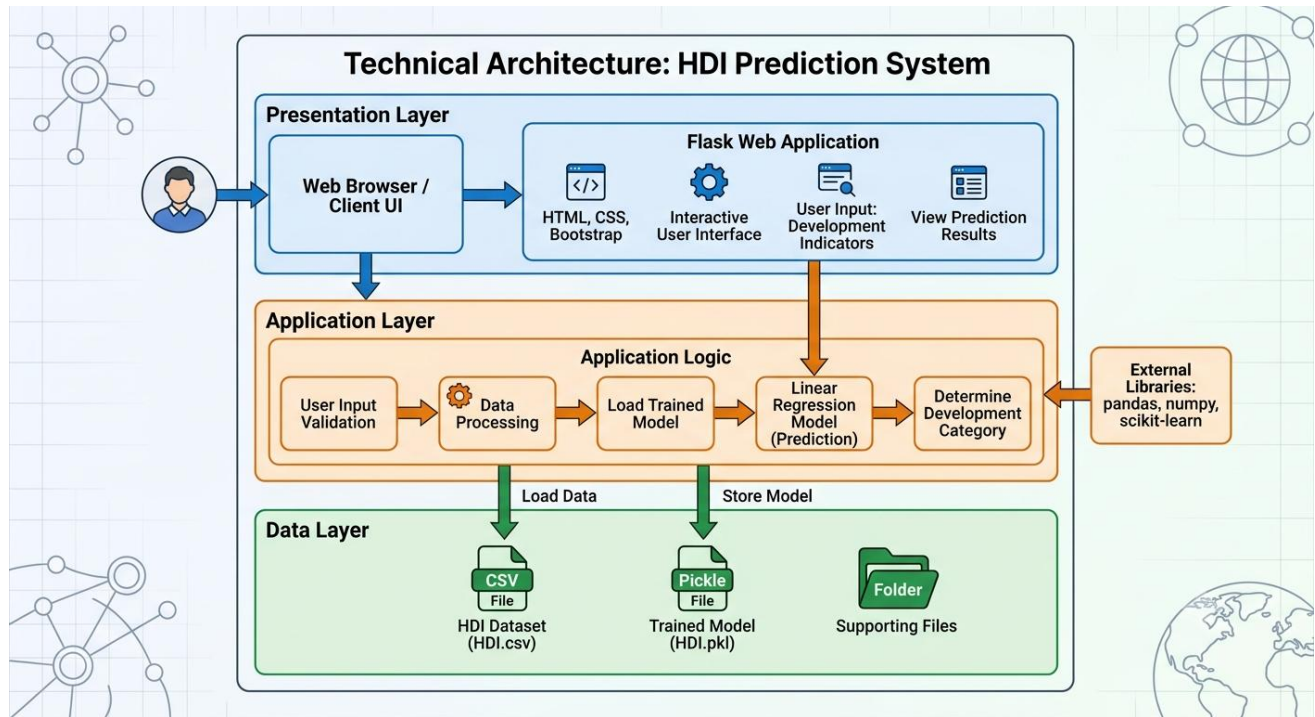
Scenario 2: Academic Research

A student or researcher uses the application to compare the development levels of different countries by entering their socio-economic indicators. The system predicts the HDI score and allows users to study the relationship between education, income, health, and overall human development.

Scenario 3: Public Awareness

An individual interested in understanding the development status of a country enters the required indicators into the application. The predicted HDI score and development category help users gain a better understanding of the country's socio-economic progress in a simple and interactive manner.

Technical Architecture



Description:

The **Human Development Index (HDI) Prediction System** follows a **three-layer architecture** consisting of the **Presentation Layer**, **Application Layer**, and **Data Layer**.

The **Presentation Layer** is a Flask-based web application developed using **HTML, CSS, and Bootstrap**, providing an interactive user interface where users can enter development indicators and view prediction results.

The **Application Layer** validates the user inputs, processes the data, loads the trained **Linear Regression** model, predicts the Human Development Index (HDI), and determines the corresponding development category.

The **Data Layer** stores the HDI dataset (**HDI.csv**), the trained machine learning model (**HDI.pkl**), and the supporting files required for prediction. This layer serves as the primary source for model training and prediction.

Pre-requisites

- Python Programming
- Flask Framework
- Machine Learning using Scikit-learn
- Linear Regression
- Pandas
- NumPy
- Pickle
- HTML
- CSS

- Bootstrap
- Visual Studio Code (VS Code)
- Git and GitHub

Project Workflow

The development of the **Human Development Index (HDI) Prediction System** is organized into five major milestones. Each milestone represents a specific stage of development, beginning with dataset preparation and ending with application testing and deployment readiness.

Milestone 1: Dataset Preparation and Environment Setup

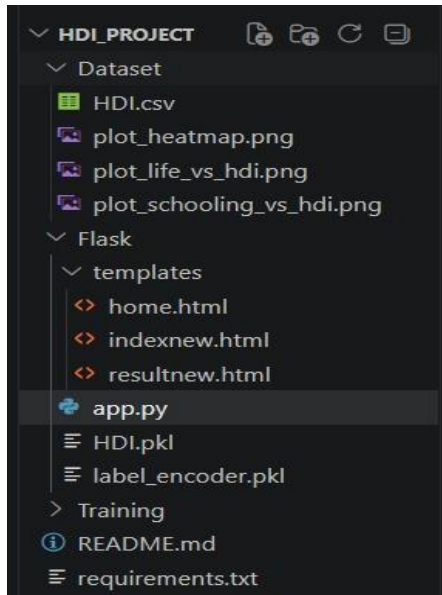
Activity 1.1: Prepare the HDI Dataset

The first step involved collecting and preparing the Human Development Index dataset required for machine learning model development. The dataset contains important socio-economic indicators including Country, Life Expectancy, Mean Years of Schooling, Gross National Income (GNI) per Capita, Internet Users, and HDI Score. The dataset was stored as HDI.csv, which serves as the primary data source for model training and prediction.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	ISO3	Region	Country	Human De	HDI	Life expec	Mean year	GNI per ca	Feature8	Feature9	Feature10	Feature11	Feature12	Feature13	Feature14	Feature15	Feature16	Feature17	Feature18	Feature19	Feature20	Feature21	Feature22
2	AFG	Region	Afghanistan	Medium	0.532	62.6	4.5	2484	26.84	53.2	80.39	97.18	81.08	23.86	98.38	81.06	73.28	32.27	93.49	11.57	67.77	12.01	65.34
3	ALB	Region	Albania	Medium	0.536	57.2	6.5	3635	61.4	44.66	51.85	47.62	86.14	48.79	42.26	51.72	77.41	70.67	46.95	76.67	12.23	5.85	9.82
4	ALG	Region	Algeria	Medium	0.538	57.3	6.1	4698	73.56	73.12	85.7	31.03	9.58	63.41	98.35	35.11	80.56	46.85	42.56	21.57	37.49	96.62	1.62
5	ARG	Region	Argentina	Medium	0.59	64.6	7.5	7626	47.29	97.62	55.89	75.85	56.66	30.42	58.56	70.4	8.89	41.46	88.95	33.73	88.8	92.03	49.35
6	AUS	Region	Australia	Medium	0.609	63.4	9.4	8519	2.49	45.62	39.89	2.22	7.18	58.6	87.47	70.95	71.35	51.81	31.81	19.16	84.12	22.54	12.34
7	BAN	Region	Bangladesh	Medium	0.636	66.6	10.1	10419	80.64	23.07	0.81	79.85	90.65	22.19	79	2.08	92.49	82	30.57	99.21	58.17	45.44	87.25
8	BRA	Region	Brazil	Medium	0.659	66.2	11.4	13060	26.57	7.44	61.91	48.89	96.83	11.05	16.45	18.84	38.52	42.72	12.39	83.08	25.36	64.74	90.61
9	CAN	Region	Canada	Medium	0.685	70.6	11.7	14185	15.02	79.1	74.37	32.4	43.59	54.84	44.47	76.27	35.28	30.16	57.28	94.37	53.04	13.54	73.71
10	CHI	Region	China	Medium	0.693	64.4	13.8	16362	41.33	87.49	86.86	67.4	78.59	40.56	84.6	96.42	44.94	44.83	98.25	16.63	65.88	83.02	73.5
11	FRA	Region	France	Medium	0.73	72.9	13.8	18456	77.35	89.16	30.88	34.3	71.28	64.31	67.14	26.61	87.76	7.63	76.57	33.27	14.05	7.5	50.11
12	GER	Region	Germany	Medium	0.745	73	14.2	19575	50.51	43.03	43.62	31.22	90.58	48.84	26.84	6.54	73.74	95.49	86.1	45.18	99.19	18.04	5.54
13	IND	Region	India	Medium	0.755	67.4	15.8	22300	43.91	86.97	41.57	89.18	22.93	58.81	1.85	69	86.16	90.49	76.21	2.8	60.98	94.41	63.05
14	IND	Region	Indonesia	Medium	0.674	73	5.4	24057	22.41	15.08	25.74	54.73	80.49	94.22	16.33	42.81	93.34	43.66	58.07	65.34	62.34	59.64	7.5
15	ITA	Region	Italy	Medium	0.7	77.9	5.1	24548	45.41	37.96	79.98	74.16	43.14	49.68	38.29	72.02	43.44	25.06	14.49	61.68	18.69	32.63	75.82
16	JAP	Region	Japan	Medium	0.707	71.6	7.3	26578	68.3	73.13	67.51	97.35	88.05	91.84	35.32	3.45	88.74	60.03	70.06	74.4	33.63	10.92	51.92
17	KEN	Region	Kenya	Medium	0.722	73.5	7.2	28909	29.63	6.1	39.25	46.93	20.25	52.47	90.44	83.56	29.87	58.8	55.79	60.22	17.41	35.42	18.84
18	MEX	Region	Mexico	Medium	0.746	73.2	8.6	30638	25.33	5.59	86.93	43.45	60.92	11.61	27.77	97.52	30.85	17.93	82.8	32.76	85.82	19.04	91.83
19	NEP	Region	Nepal	Medium	0.785	79.3	10	31769	49.31	18.37	24.03	66.33	6.58	56.53	66.62	2.36	87.56	1.39	46.33	94.17	81.51	84.63	47.68
20	NIG	Region	Nigeria	Medium	0.823	81.2	11.7	34166	32.72	15.55	62.5	1.96	36.46	62.38	81.24	60.31	5.05	60.12	89.3	78.72	16.66	17.92	50.08
21	NOR	Region	Norway	Medium	0.829	79	12.8	35235	49.68	11.03	76.82	64.89	24.24	66.47	66.25	71.12	73.32	60.54	84.69	44.11	81.29	59.84	51.77
22	PAK	Region	Pakistan	Medium	0.771	57.1	13.1	37819	16.1	90.13	3.59	11.17	6.92	29.37	25.67	42.75	46.15	59.6	26.93	64.12	50.82	15.67	30.83
23	RUS	Region	Russia	Medium	0.791	57.6	14.6	39680	96.46	85.38	17.4	23.32	75.61	47.38	17.69	96.29	59.6	1.97	2.4	17.19	72.97	70.2	47.44
24	SOU	Region	South Africa	Medium	0.832	62.9	15.2	42052	88.54	94.46	64.83	53.33	60.86	80.8	24.12	32.7	91.6	98.26	97.72	37.56	37.54	42.41	70.21
25	SPA	Region	Spain	Medium	0.848	62.8	16.3	43344	65.85	58.6	11.59	33	5.07	63.27	84.01	25.79	29.66	20.2	99.01	50.39	53.3	32.66	34.74
26	SRI	Region	Sri Lanka	Medium	0.753	64.3	4.8	45687	78.33	72.37	79.06	41.42	79.64	45.47	62.3	34.21	61.04	29.69	16.53	28	90.47	98.82	34.57

Activity 1.2: Configure the Development Environment

The development environment was configured using Python and Visual Studio Code (VS Code). All required libraries, including Flask, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, and Pickle, were installed using the **requirements.txt** file to ensure a consistent project environment.



Activity 1.3: Inspect and Understand the Dataset

The dataset was loaded using the Pandas library to verify its structure, feature names, data types, and overall quality. Descriptive statistics and basic data analysis were performed to understand the relationships between development indicators before training the model.

```
PS C:\Users\sp\Documents\HDI_Project> python Training/HumDevIndex.py
Dataset shape: (30, 68)
  ISO3  Region  Country Human Development Group  ...  Feature64  Feature65  Feature66  Internet users
0  AFG  Region  Afghanistan                Medium  ...    30.07    15.07    88.77             8.0
1  ALB  Region  Albania                  Medium  ...    29.72    84.34    79.10             8.0
2  ALG  Region  Algeria                  Medium  ...    82.08    51.52    18.82             9.5
3  ARG  Region  Argentina                Medium  ...    81.60    74.99    43.51            13.7
4  AUS  Region  Australia                Medium  ...    32.19    49.34    30.17            14.6

[5 rows x 68 columns]

Column names:
['ISO3', 'Region', 'Country', 'Human Development Group', 'HDI', 'Life expectancy', 'Mean years of schooling', 'GNI per
capita', 'Feature8', 'Feature9', 'Feature10', 'Feature11', 'Feature12', 'Feature13', 'Feature14', 'Feature15', 'Featu
re16', 'Feature17', 'Feature18', 'Feature19', 'Feature20', 'Feature21', 'Feature22', 'Feature23', 'Feature24', 'Featur
e25', 'Feature26', 'Feature27', 'Feature28', 'Feature29', 'Feature30', 'Feature31', 'Feature32', 'Feature33', 'Feature
34', 'Feature35', 'Feature36', 'Feature37', 'Feature38', 'Feature39', 'Feature40', 'Feature41', 'Feature42', 'Feature4
3', 'Feature44', 'Feature45', 'Feature46', 'Feature47', 'Feature48', 'Feature49', 'Feature50', 'Feature51', 'Feature52
', 'Feature53', 'Feature54', 'Feature55', 'Feature56', 'Feature57', 'Feature58', 'Feature59', 'Feature60', 'Feature61
', 'Feature62', 'Feature63', 'Feature64', 'Feature65', 'Feature66', 'Internet users']

Unique countries: 30
<StringArray>
[ 'Afghanistan',      'Albania',      'Algeria',      'Argentina',
  'Australia',      'Bangladesh',  'Brazil',      'Canada',
  'China',          'France',      'Germany',      'India',
  'Indonesia',      'Italy',       'Japan',       'Kenya',
  'Mexico',         'Nepal',      'Nigeria',     'Norway',
  'Pakistan',       'Russia',     'South Africa', 'Spain',
  'Sri Lanka',     'Sweden',     'Switzerland', 'United Kingdom',
  'United States',  'Vietnam']
Length: 30, dtype: str
```

Activity 1.4: Design the System Architecture

A three-layer architecture consisting of the Presentation Layer, Application Layer, and Data Layer was designed to separate the user interface, prediction logic, and data storage, making the application modular and easy to maintain.

Milestone 2: Data Preprocessing

Activity 2.1: Data Preparation

The HDI dataset was loaded into the machine learning pipeline. The required independent variables (Life Expectancy, Mean Years of Schooling, GNI per Capita, Internet Users) and the dependent variable (HDI Score) were selected for model training.

```
x shape: (30, 5)
y shape: (30, 1)

Null values in x:
Country          0
Life expectancy  0
Mean years of schooling  0
GNI per capita   0
Internet users   0
dtype: int64

Null values after filling:
Country          0
Life expectancy  0
Mean years of schooling  0
GNI per capita   0
Internet users   0
dtype: int64

Training samples: 27
Testing samples: 3

Model trained successfully.
```

Activity 2.2: Train-Test Split

The dataset was divided into training and testing datasets using the **train_test_split()** function to evaluate the performance of the Linear Regression model on unseen data.

```
# Step 8: Train / Test Split (90% train, 10% test)
x_train, x_test, y_train, y_test = train_test_split(
    X, y, test_size=0.1, random_state=42
)
print("\nTraining samples:", x_train.shape[0])
print("Testing samples:", x_test.shape[0])
```

Activity 2.3: Data Verification

The selected features were verified for consistency, and the training data was prepared for Linear Regression model development.

```
# Step 6: Handle Null Values
print("\nNull values in X:")
print(X.isnull().sum())

X.fillna(X.mean(numeric_only=True), inplace=True)

print("\nNull values after filling:")
print(X.isnull().sum())

# Step 7: Label Encoding for Country column
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
X["Country"] = le.fit_transform(X["Country"].astype(str))
with open("Flask/label_encoder.pkl", "wb") as f:
    pickle.dump(le, f)
```

Milestone 3: Machine Learning Model Development

Activity 3.1: Train Multiple Machine Learning Models

The Linear Regression algorithm was trained using the prepared HDI dataset to learn the relationship between socio-economic indicators and the Human Development Index.

```
# Step 9: Fit Linear Regression Model
reg = LinearRegression()
reg.fit(x_train, y_train)
print("\nModel trained successfully.")
```

Activity 3.2: Model Evaluation

The trained model was evaluated using performance metrics including:

- R^2 Score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)

The evaluation confirmed that the model accurately predicts the Human Development Index.

```
Model trained successfully.
```

```
Predicted values:
```

```
[[0.80682275]  
 [0.72236222]  
 [0.84817735]]
```

```
R2 Score: 0.9999
```

```
Sample prediction: 0.59
```

Activity 3.3: Save the Trained Model

The trained Linear Regression model was saved as HDI.pkl using the Pickle library. This model is loaded by the Flask application during prediction.

```
# Step 11: Save Model as Pickle  
with open("Flask/HDI.pkl", "wb") as f:  
    pickle.dump(reg, f)  
  
print("\nModel saved as Flask/HDI.pkl – ready for deployment!")
```

Milestone 4: Flask Web Application Development

Activity 4.1: Develop the Flask Backend

The backend application was developed using Flask. It receives user inputs, loads the trained Linear Regression model, predicts the Human Development Index, and displays the prediction result.

```
> app.py > home  
import numpy as np  
import pandas as pd  
from flask import Flask, render_template, request  
import pickle  
  
app = Flask(__name__)  
  
# Load the trained model  
model = pickle.load(open("HDI.pkl", "rb"))  
  
# — Routes —  
  
@app.route("/")  
@app.route("/Home", methods=["POST", "GET"])  
def home():  
    return render_template("home.html")  
  
@app.route("/Prediction", methods=["POST", "GET"])  
def prediction():  
    return render_template("indexnew.html")  
  
@app.route("/predict", methods=["POST"])  
def predict():  
    # Read form values submitted by user  
    input_features = [float(x) for x in request.form.values()]  
    features_value = [np.array(input_features)]
```

```
app.py > home
def predict():

    features_name = [
        "Country",
        "Life expectancy",
        "Mean years of schooling",
        "GNI per capita",
        "Internet users",
    ]

    df = pd.DataFrame(features_value, columns=features_name)

    # Generate prediction
    output = model.predict(df)
    y_pred = round(output[0][0], 2)

    # Classify into HDI tier
    if 0.30 <= y_pred <= 0.40:
        prediction_text = "Low HDI - " + str(y_pred)
    elif 0.40 < y_pred <= 0.70:
        prediction_text = "Medium HDI - " + str(y_pred)
    elif 0.70 < y_pred <= 0.80:
        prediction_text = "High HDI - " + str(y_pred)
    elif 0.80 < y_pred <= 0.94:
        prediction_text = "Very High HDI - " + str(y_pred)
    else:
        prediction_text = (
            "The given values do not match the expected range. "
            "Please check your inputs."
        )

    return render_template("resultnew.html", prediction_text=prediction_text)

# — Entry Point —

if __name__ == "__main__":
    app.run(debug=True, port=5000)
```

Activity 4.2: Develop the User Interface

A responsive web interface was developed using HTML, CSS, and Bootstrap.

The application includes:

- Home Page
- Prediction Page
- Result Page

The interface allows users to enter development indicators and instantly view the predicted HDI score.

HDI Predictor

Human Development Index

A machine learning model that predicts a country's HDI score using health, education, and income indicators — powered by Linear Regression.

Predict HDI →

Very High ≥ 0.80 High $0.70 - 0.79$ Medium $0.40 - 0.70$ Low ≤ 0.40



Health

Life expectancy at birth reflects the overall quality of healthcare and living conditions.



Education

Mean and expected years of schooling measure access and quality of education systems.



Income

Gross National Income (GNI) per capita (PPP \$) reflects the standard of living.

The Human Development Index was created by the United Nations Development Programme to emphasize that people and their capabilities are the ultimate criteria for assessing a country's development — not

Predict HDI Score

Enter development indicators to get the predicted HDI classification.

QUICK FILL

NO Norway IN India NE Niger DE Germany

Country

Australia

Used as encoded numeric feature in the model

Life expectancy (years)

73

Range: 40 – 90 years

Mean years of schooling

8

Average years of education completed (0 – 15)

Gross national income (GNI) per capita (PPP \$)

4000

Range: \$500 – \$100,000

Internet users (%)

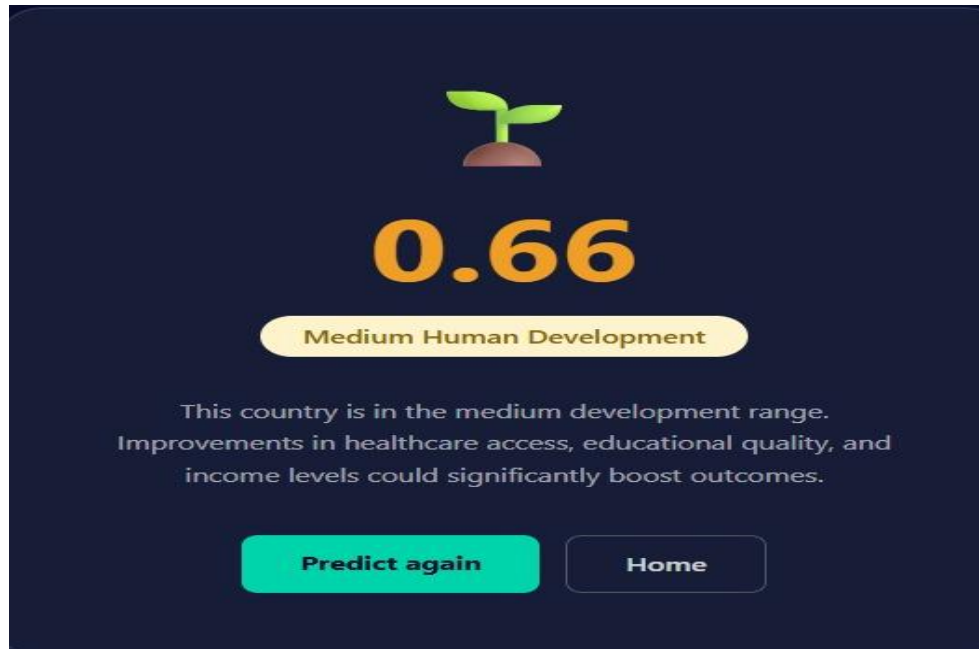
42

Percentage of population using internet (0 – 100)

Predict HDI →

Activity 4.3: Integrate the Machine Learning Model

The trained Linear Regression model was integrated with the Flask application. After the user submits the required development indicators, the application predicts the Human Development Index and displays the corresponding development category.



Milestone 5: Testing and Deployment Preparation

Activity 5.1: Local Application Testing

The application was executed locally using Flask. Different combinations of development indicators were tested to verify that the prediction results were generated correctly.

Activity 5.2: Prediction Verification

Multiple test cases were executed to verify the consistency and accuracy of the prediction results. The application successfully displayed the predicted HDI score and development category for different input values.

Activity 5.3: Deployment Preparation

The project files were organized for deployment. Essential files including **HDI.csv**, **HDI.pkl**, **requirements.txt**, **README.md**, HTML templates, static files, and Flask source code were included. The application was successfully tested on the local system and prepared for deployment.